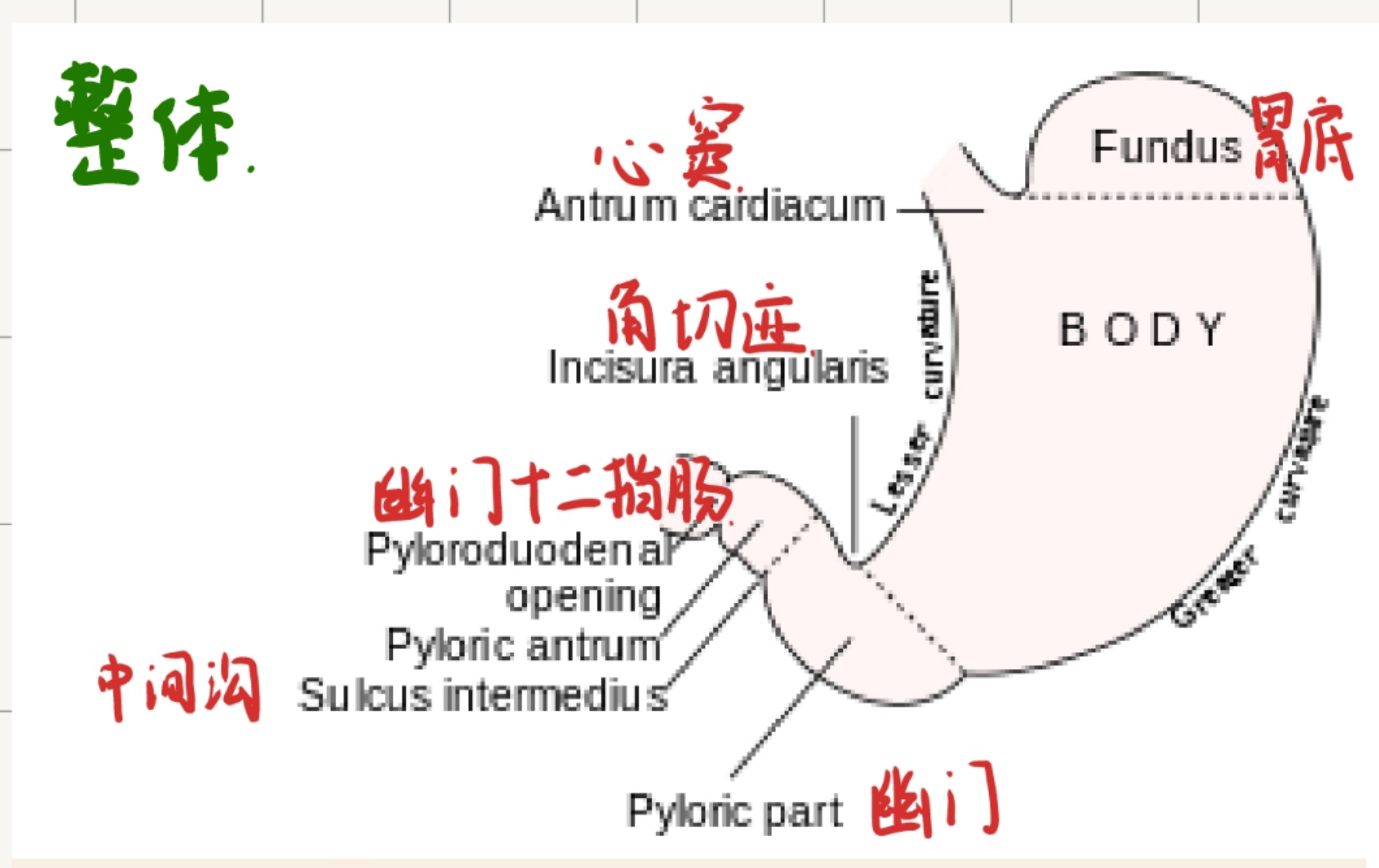


消化器官(胃.肠.肝)

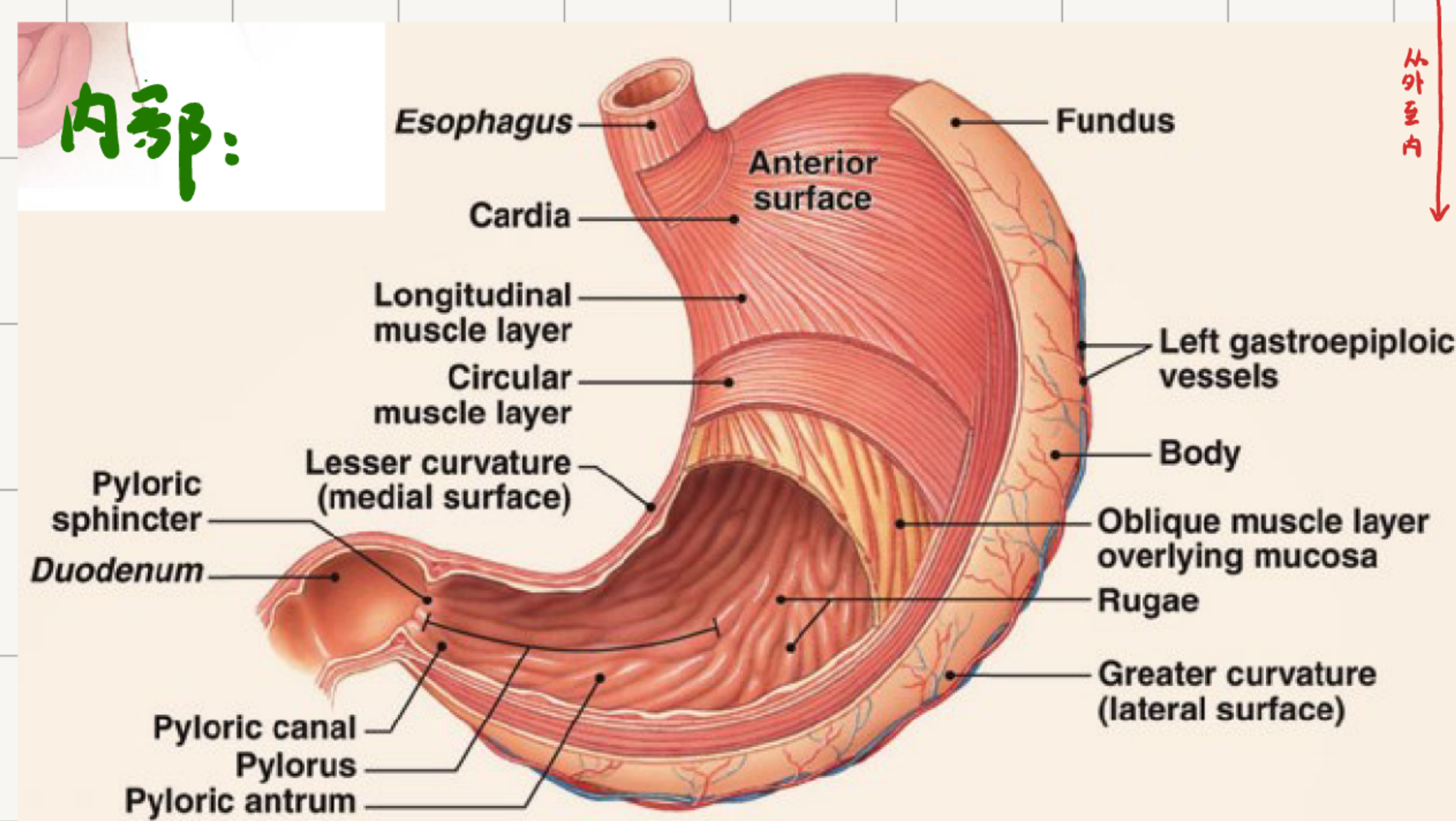
1. 胃

结构: 整体



血供: { gastric artery (branch of celiac artery)
gastric vein to portal vein.
肠神经丛: 广泛分布在消化系统, 独立于交感/副交感 (enteric nervous system) 有 myenteric, submucosa 两层.

内部:



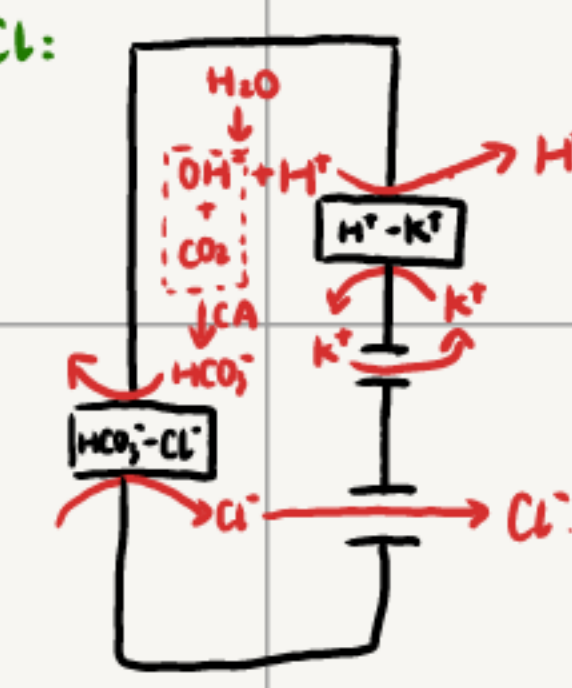
从外至内
serosa 浆膜
myenteric 肌层 { longitudinal muscle 纵肌
circular 环肌
oblique 斜肌 } myenteric plexus
submucosa ← submucosa plexus
mucosa { muscularis muscle 平滑肌
rugae { gastric gland
gastric pit
细胞类型: mucous neck cell: 分泌 mucus, HCO₃⁻
chief cell: { pepsinogen
gastric lipase
D cell: somatostatin. 抑制 H⁺
G cell: gastrin. 促进 H⁺ (enteroendocrine)
Enterochromaffin cell: 组胺 (histamine)
parietal cell { HCl:
intrinsic factor.

功能: food storage in fundus.

机械消化. (body, antrum)

分泌胃液 (gastric juice): { HCl, pepsinogen, IF
produce chyme. 双层: 形成 pH 梯度. 保护.

protection: 为什么胃不会自我消化
① 上皮 cell 间 tight junction
② surface and neck mucus.
③ PGE: 促进 mucus 和 HCO₃⁻ 分泌.



调节: ^{激素/神经} somatostatin $\rightarrow$ SSIR $\rightarrow$ cAMP { parietal cell (H^+)
(D cell) { ECL (histamine)
G (gastrin)

② gastrin $\rightarrow$ CCK2 $\rightarrow$ Ca^{2+} { parietal (H^+)
(G cell) $\rightarrow$ 促胰液素 { ECL (histamine)
肠. 胃

③ histamine $\rightarrow$ H_2 $\rightarrow$ Ca^{2+} /cAMP $\rightarrow$ parietal (H^+)
(ECL)

① ACh $\rightarrow$ M_3 $\rightarrow$ Ca^{2+} $\rightarrow$ { ECL (histamine)
(vagal. enteronerve) { parietal (H^+)
G cell (gastrin)

$H^+ \rightarrow$ somatostatin.

时期:

{ cephalic (头期): 迷走激活 ECL. gastrin. 促 H^+ .

{ gastric (胃期): ^(enteric neuron reflexes) { 局部反射 $\rightarrow$ ACh $\rightarrow$ H^+
食物中 peptide $\rightarrow$ gastrin $\rightarrow$ H^+ { 碱性 $\rightarrow$ H^+
HCl $\rightarrow$ somatostatin $\rightarrow$ H^+ .

{ intestinal (肠期) { CCK, secretin, GLP, GIP $\rightarrow$ { histamine
gastrin.

neural reflex $\rightarrow$ ACh.

HCl $\rightarrow$ somatostatin $\rightarrow$ H^+ .

2. 肠/胰

肠:

结构: duodenum ← sphincter of Oddi → common bile duct

pancreatic duct

jejunum

ileum

① 控制胆汁、胰液流入

② 防止 duodenal reflux

开关受 CCK、nerve 调控

从外至内

serosa

muscularis externa { outer longitudinal muscle
inner circular muscle.

submucosa: Brunner's gland, 分泌 alkaline mucus.

mucosa { villi: enterocyte (有 microvilli), goblet cell

crypts of Lieberkuhn: 肠腺, 包含 f cell. Paneth (release antimicrobial mediator)

Lamina propria

muscularis mucosae 黏膜肌

功能:

肠液: { mucus: { single, penetrable: 保护 + 允许 chyme 贴近

调节 pH, 便于 enzyme 作用.

goblet cell, Brunner's gland

Amp: 抗菌 paneth

水, 盐: enterocyte in crypt

消化酶: 全部位于 enterocyte 表面或 cell 内

{ disaccharidase
enteropeptidase
aminopeptidase
lipase

内分泌: EEC进行, 兼有感受、释放 (crypt and villi)

secretin (S type 分泌): 促胰液素

pH↓, secretin↑, 促胰分泌 HCO_3^- , 抑胃排空.

CCK (I type): 缩胆囊素, 促胰酶素

FA, aa↑, 迷走↑, CCK分泌, 作用于CCK_A, CCK_B, 与CCK₂受体区分!
促胆囊收缩, sphincter of Oddi放松, 促胰acinar cell分泌酶

gastrin (G type): 促胃 H^+

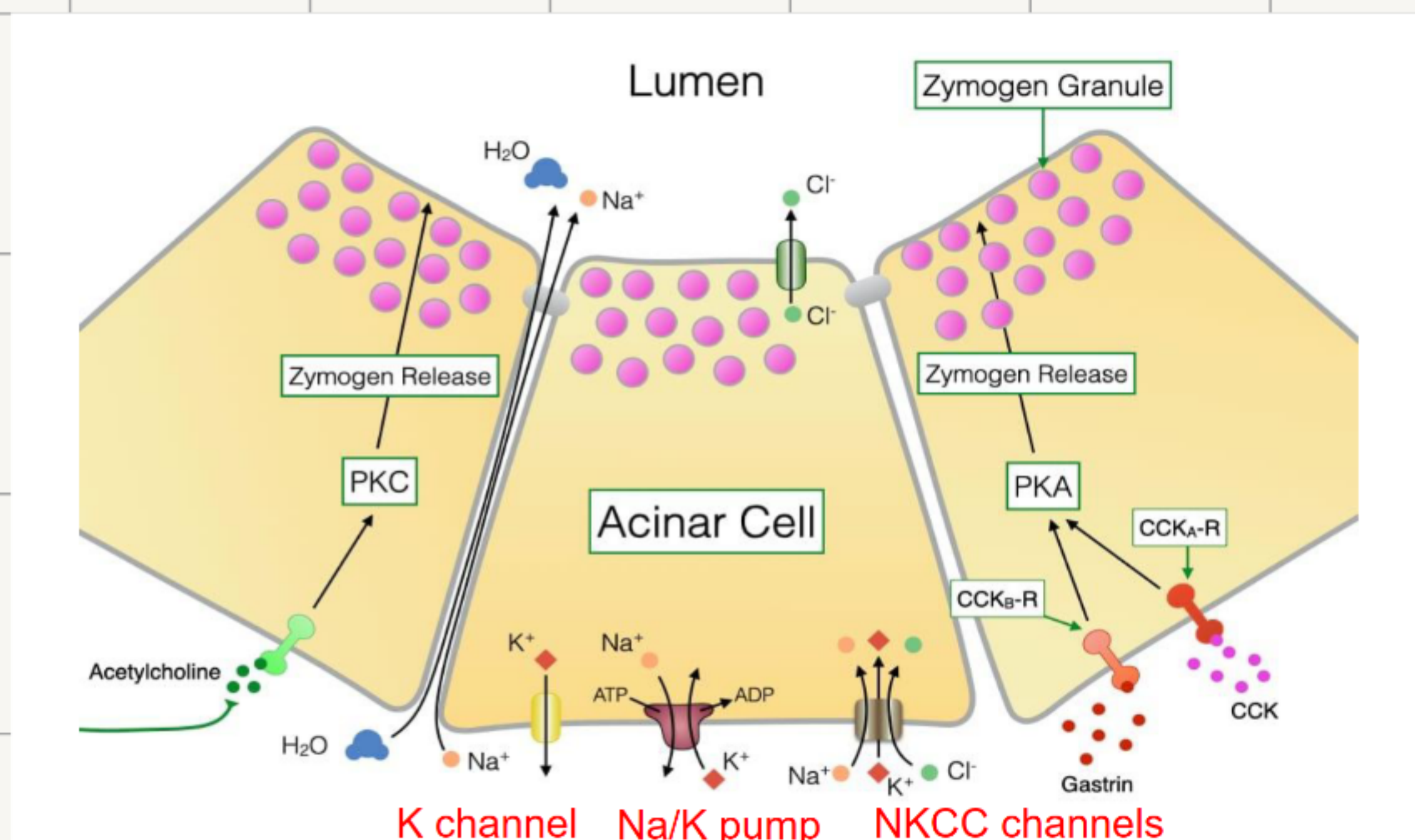
GLP (L type)

GIP (K type)

调控: acidic/hypertonic chyme → 肠
vagal → 肠
CCK, secretin → 液
交感 → 液

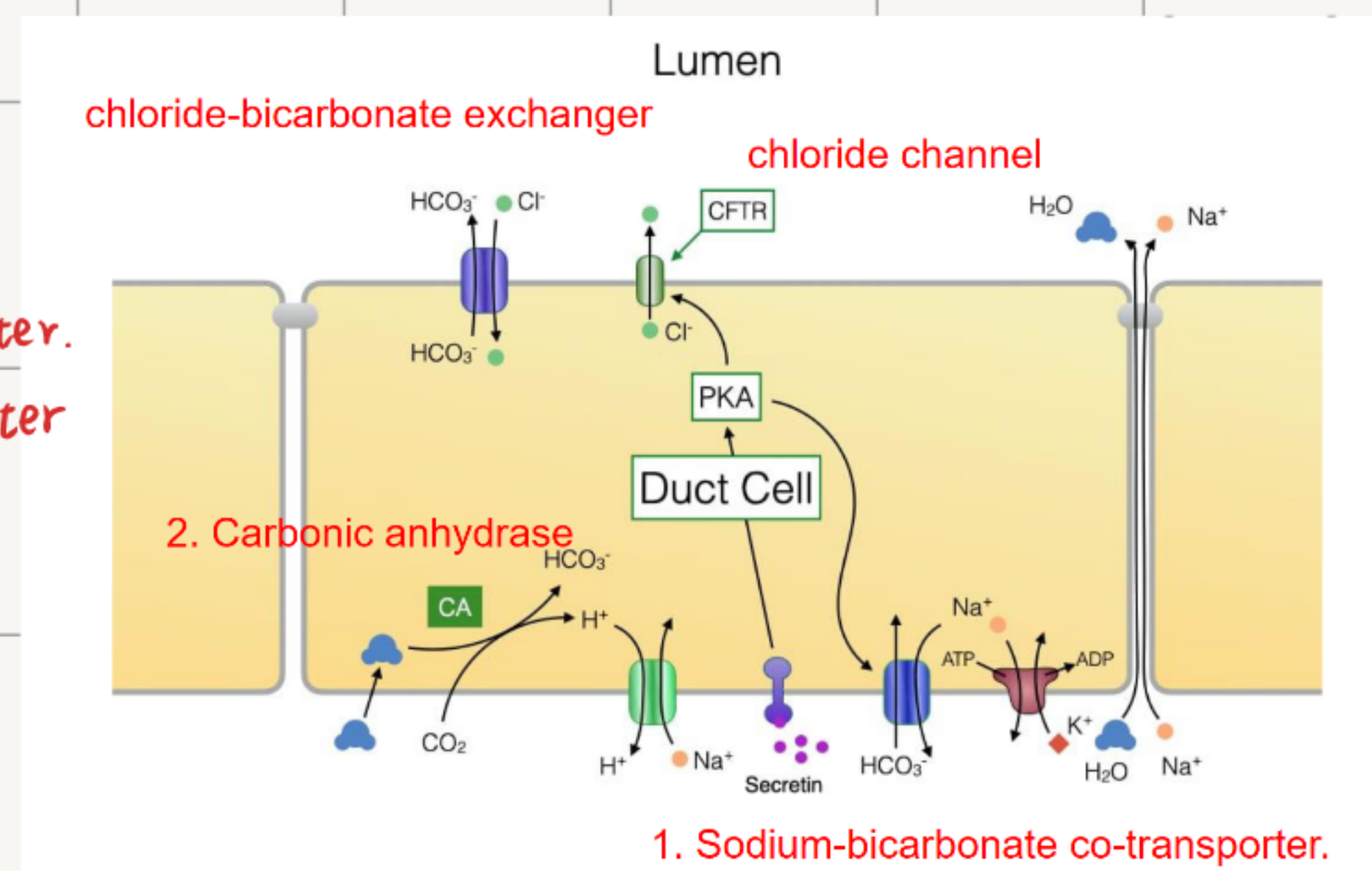
胰结构 { acinar cell

{ zymogen: 受Gq调控
NaCl: Na^+ - K^+ , K^+ , NKCC.



duct cell.

apical: { CFTR
 Cl^- - HCO_3^- antiporter.
basal: { Na^+ - HCO_3^- antiporter
CA.



pancreatic juice: 酶 + HCO_3^- / NaCl electrolytes.

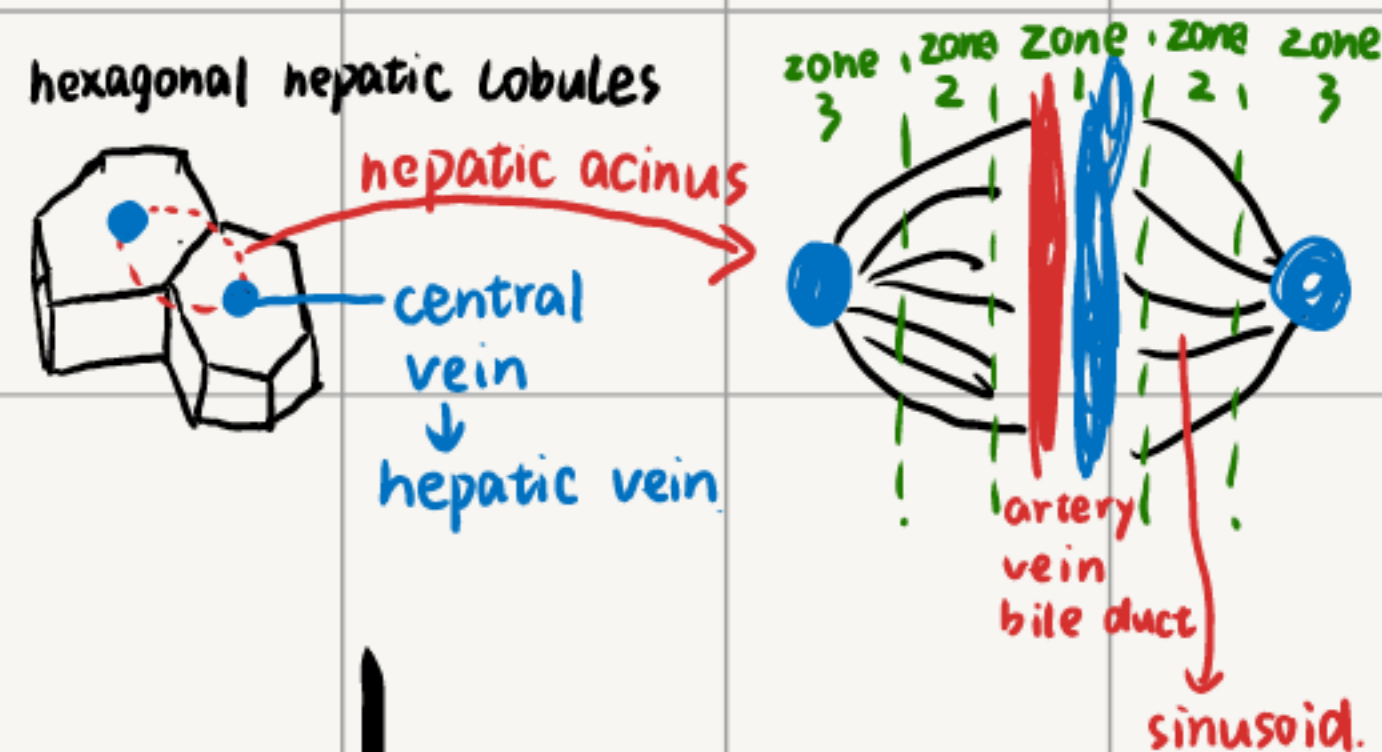
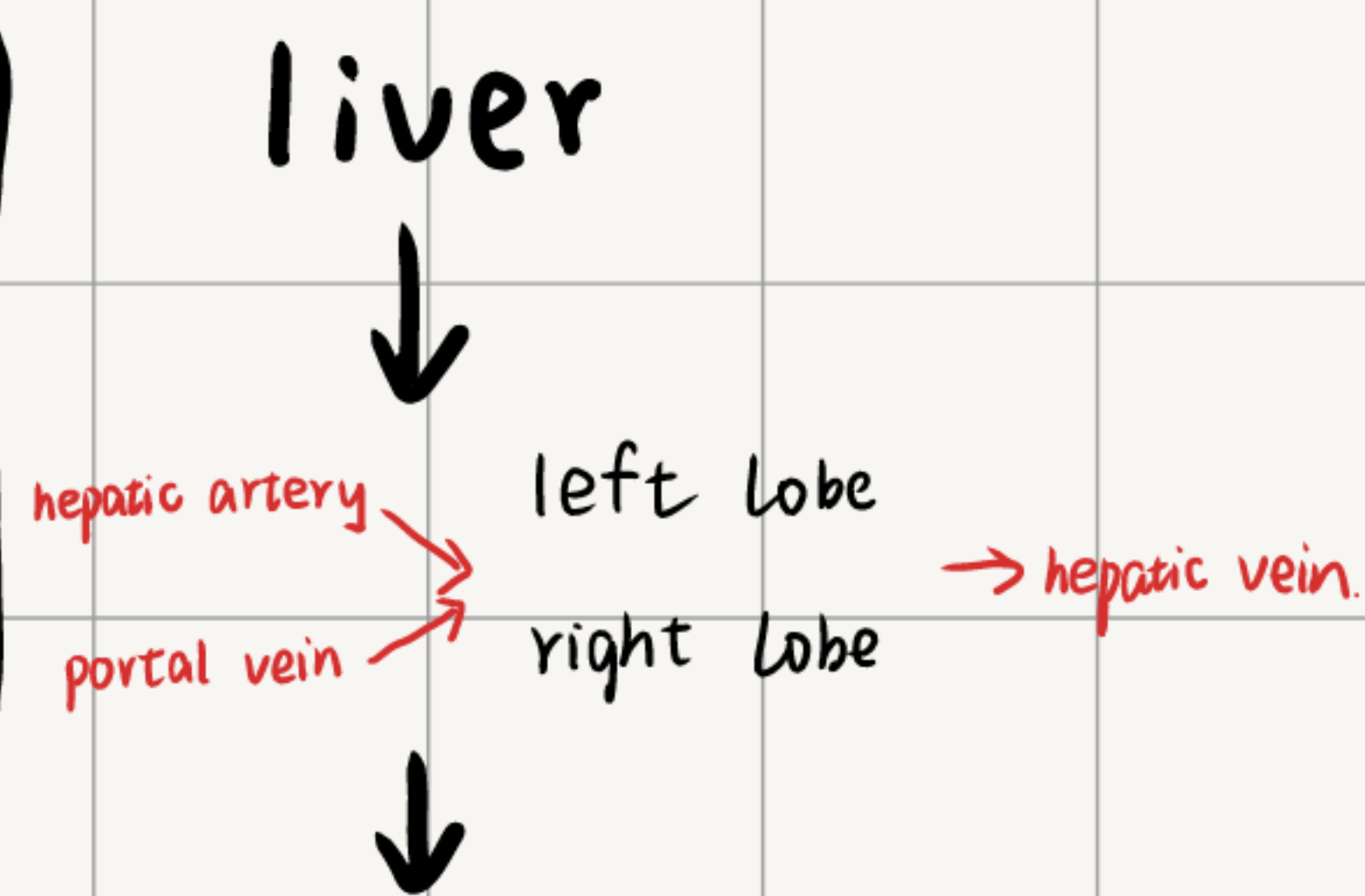
调控: secretin, CCK

{ cephalic: vagal.
gastric: gastrin促进.
intestinal: 主要.

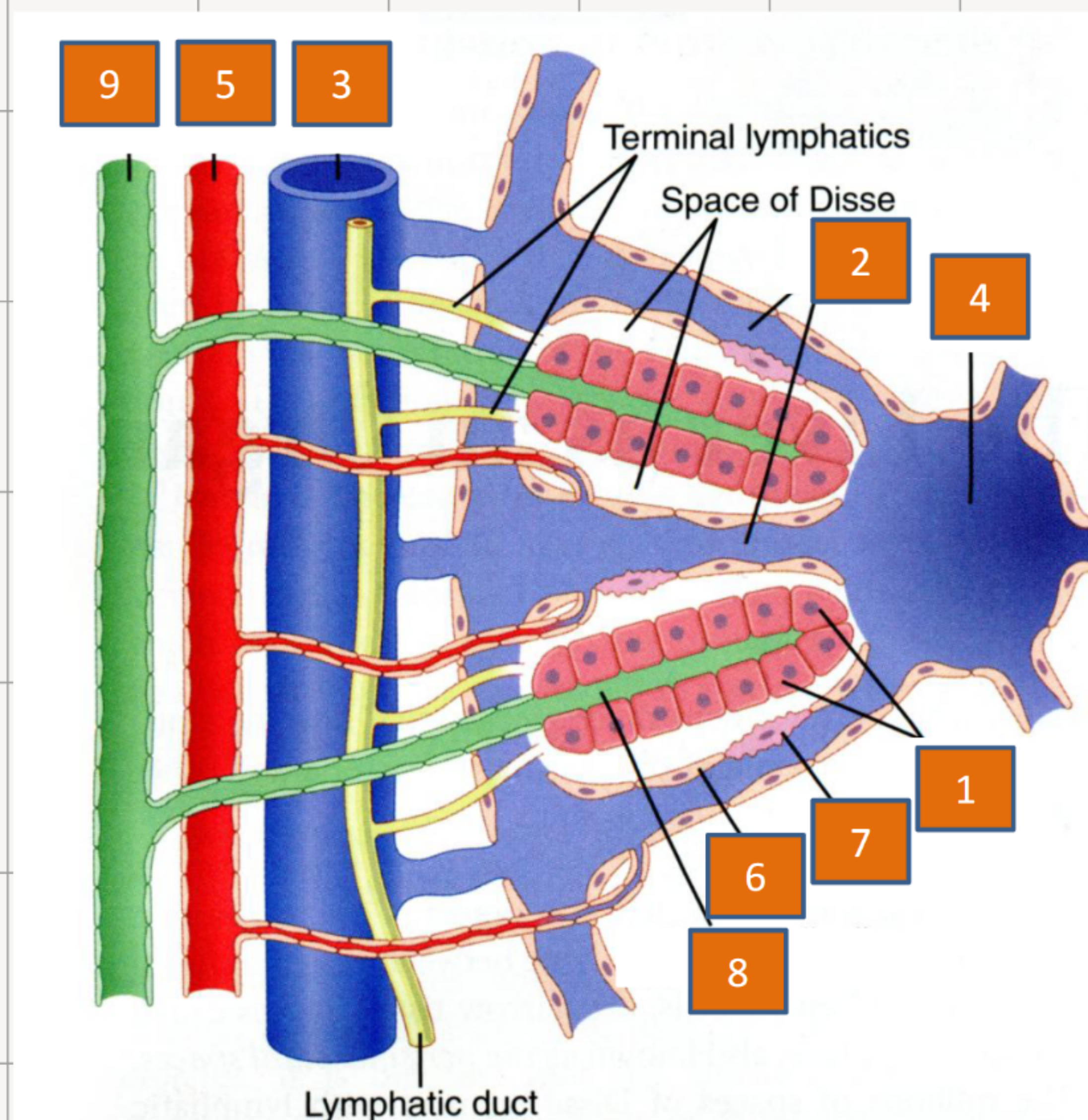
3. 肝

结构:

从大到小



zone 1 (periportal): 首个接受O₂, 营养
丰富, 营养多, 循环好.
zone 2 (middle)
zone 3 (centrilobular): 循环差, 胆管阻塞不明显.



1. hepatocyte: { 主要功能
finger-like projection.
3. portal vein
5. hepatic artery
2. sinusoid: highly permeable capillaries
4. central vein → hepatic vein.
6. sinusoidal endothelial cell: { sieve-like plate: 用于筛大分子.
无基膜, 不紧密, 便于肝cell接触血.
7. Kupffer: phagocytes, 除去衰老WBC, RBC, 细菌, 外来物质.
8. bile canaculi: 与 hepatocyte 相连, 运输胆汁
9. bile duct.

功能:

① bile secretion

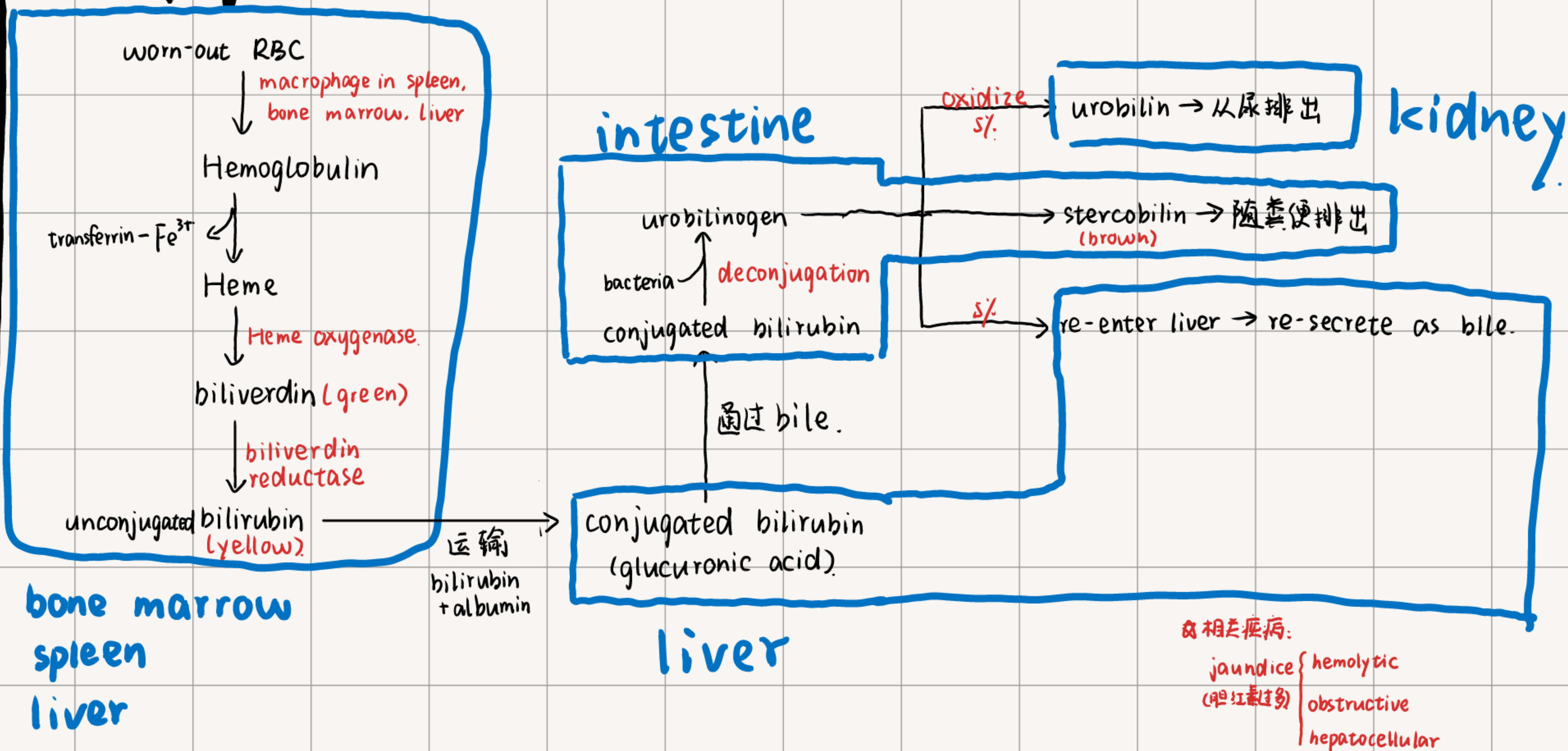
肝分泌

bile salt: 胆固醇衍生物. 以 ^{cholic acid} CA, ^{chenodeoxycholic acid} CDCA conjugated with Gly 被分泌, 可被 microbiome 再处理

★具备 enterohepatic 循环

肝分泌 bile salt
↓
肠利用 Na^+ -anion transporter, active bile salt transporter 吸收
↓
血中 bile salt

bile pigment: 以 bilirubin 为例.



cholesterol, phospholipids, mucus

胆管上皮细胞分泌: H_2O , HCO_3^-

功能: 排出胆色素, detergent in lipid absorption, form micelles.

调控: bile salt, CCK, vagal 促胆汁.

② 蛋白合成 { serum albumin: ^{60% concentration + 40% retention of ion.} oncotic pressure + transport
globulin (α, β): 例如 SHBG
prothrombin, fibrinogen.

③ 内分泌 { IGF-1: GH作用/paracrine factor, 促生长, 部分起insulin作用
(肝、肾、骨髓造血分泌)
thrombopoietin: 组成型分泌, 促 megakaryocytes

④ detoxification.

Phase I { oxidation,
reduction

⇒ { phosphatidylcholine (PC): 电子转移
cytochrome P450: 接受电子

hydrolysis

Phase II: conjugation to glucuronic acid
⇒ 经肾排出.

